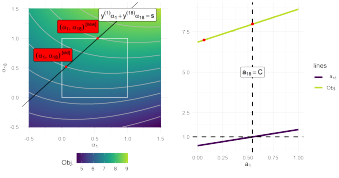
# Introduction to Machine Learning

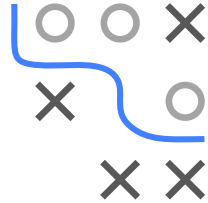
# Linear Support Vector Machines
# Support Vector Machine Training




**Learning goals**

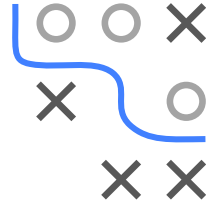
- Know that the SVM problem is not differentiable
- Know how to optimize the SVM problem in the primal via subgradient descent
- Know how to optimize SVM in the dual formulation via pairwise coordinate ascent

# SUPPORT VECTOR MACHINE TRAINING



- Until now, we have ignored the issue of solving the various convex optimization problems.
- The first question is whether we should solve the **primal** or the **dual problem**.
- In the literature SVMs are usually trained in the dual.
- However, SVMs can be trained both in the primal and the dual – each approach has its advantages and disadvantages.
- It is not easy to create an efficient SVM solver, and often specialized appraoches have been developed, we only cover basic ideas here.

# TRAINING SVM IN THE PRIMAL



Unconstrained formulation of soft-margin SVM:

$$\min_{\boldsymbol{\theta},\theta_0} \quad \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2 + \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

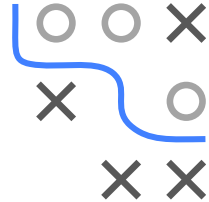
where $L(y, f) = \max(0, 1 - yf)$ and $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^T\mathbf{x} + \theta_0$.
(We inconsequentially changed the regularization constant.)

We cannot directly use GD, as the above is not differentiable.

**Solutions:**

1. Use smoothed loss (squared hinge, huber), then do GD.
   NB: Will not create a sparse SVM if we do not add extra tricks.
2. Use **subgradient** methods.
3. Do stochastic subgradient descent.
   Pegasos: Primal Estimated sub-GrAdient SOlver for SVM.

# PEGASOS: SSGD IN THE PRIMAL



Approximate the risk by a stochastic 1-sample version:

$$\frac{\lambda}{2}\|\boldsymbol{\theta}\|^2 + L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

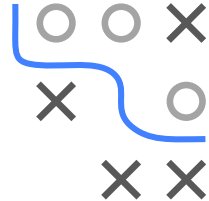
With: $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^T\mathbf{x} + \theta_0$ and $L(y, f) = \max(0, 1 - yf)$
The subgradient for $\boldsymbol{\theta}$ is $\lambda\boldsymbol{\theta} - y^{(i)}\mathbf{x}^{(i)}\mathbb{I}_{yf<1}$

Stochastic subgradient descent (without intercept $\theta_0$)

1: **for** $t = 1, 2, \ldots$ **do**
2: Pick step size $\alpha$
3: Randomly pick an index $i$
4: If $y^{(i)}f\left(\mathbf{x}^{(i)}\right) < 1$ set $\boldsymbol{\theta}^{[t+1]} = (1 - \lambda\alpha)\boldsymbol{\theta}^{[t]} + \alpha y^{(i)}\mathbf{x}^{(i)}$
5: If $y^{(i)}f\left(\mathbf{x}^{(i)}\right) \geq 1$ set $\boldsymbol{\theta}^{[t+1]} = (1 - \lambda\alpha)\boldsymbol{\theta}^{[t]}$
6: **end for**

Note the weight decay due to the L2-regularization.

# TRAINING SVM IN THE DUAL



The dual problem of the soft-margin SVM is

$$\max_\alpha \quad \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y^{(i)} y^{(j)} \left\langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \right\rangle$$

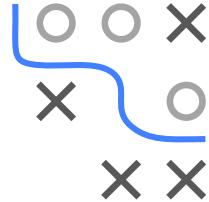
$$\text{s.t.} \quad 0 \leq \alpha_i \leq C \quad \sum_{i=1}^{n} \alpha_i y^{(i)} = 0$$

We could solve this problem using coordinate ascent. That means we optimize w.r.t. $\alpha_1$, for example, while holding $\alpha_2, \ldots, \alpha_n$ fixed.

But: We cannot make any progress since $\alpha_1$ is determined by $\sum_{i=1}^{n} \alpha_i y^{(i)} = 0$!

# TRAINING SVM IN THE DUAL / 2

**Solution:** Update two variables simultaneously



$$\max_\alpha \quad \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y^{(i)}y^{(j)} \left\langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \right\rangle$$

$$\text{s.t.} \quad 0 \leq \alpha_i \leq C \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0$$

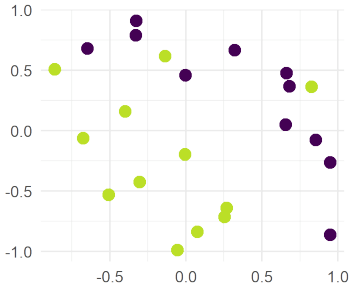
Pairwise coordinate ascent in the dual

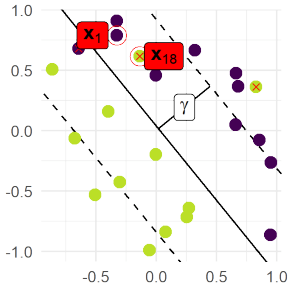
1: Initialize $\boldsymbol{\alpha} = 0$ (or more cleverly)
2: **for** $t = 1, 2, \ldots$ **do**
3: Select some pair $\alpha_i, \alpha_j$ to update next
4: Optimize dual w.r.t. $\alpha_i, \alpha_j$, while holding $\alpha_k$ ($k \neq i, j$) fixed
5: **end for**

The objective is quadratic in the pair, and $s := y^{(i)}\alpha_i + y^{(j)}\alpha_j$ must stay constant. So both $\alpha$ are changed by same (absolute) amount, the signs of the change depend on the labels.

# TRAINING SVM IN THE DUAL / 3

Assume we are in a valid state, $0 \leq \alpha_i \leq C$. Then we chose[1] two observations (encircled in red) for the next iteration. Note they have opposite labels so the sign of their change is equal.



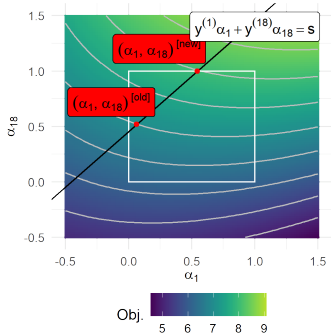

[1] There are heuristics to pick the observations to speed up convergence.

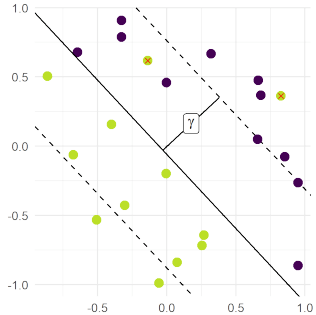
# TRAINING SVM IN THE DUAL

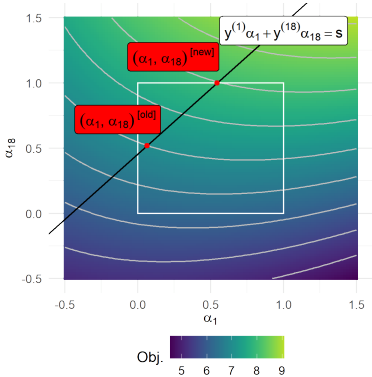
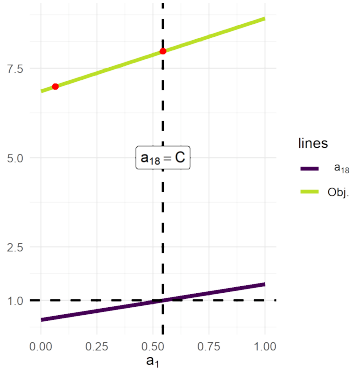
$$\max_\alpha \quad \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y^{(i)}y^{(j)}\left\langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right\rangle$$

$$\text{s.t.} \quad 0 \le \alpha_i \le C \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0$$

We move on the linear constraint until the pair-optimum or the bounday (here: $C = 1$).



# TRAINING SVM IN THE DUAL / 2

Sequential Minimal Optimization (SMO) exploits the fact that effectively we only need to solve a one-dimensional quadratic problem, over in interval, for which an analytical solution exists.